Multi-Dialect Vietnamese Text-to-Speech (TTS)

Cheston Opsasnick, Vy Bui-Nguyen, Annalise Xiao, Quenton Ni

Team Lower Expectations
University of Minnesota-Twin Cities, Minneapolis, MN

Introduction

Problem Definition & Motivation:

- In recent years, TTS systems have rapidly improved in their naturalness and intelligibility with paradigms like continuous flow matching and audio language models. However, most of these models are based on popular, high resource languages such as English and Mandarin. Vietnamese still remains underrepresented as a low-resource language.
- Previous work has focused on dialect identification and speech recognition for Vietnamese and most existing Vietnamese speech datasets were designed for Automatic Speech Recognition (ASR) and Dialect Identification (DI).
- The strict lexical tones and phonetic shifts between Northern, Central, and Southern Vietnamese present complex acoustic challenges that traditional synthesis pipelines and English-centric models struggle to resolve without dialect-aware fine tuning.

Proposed Idea:

- This project investigates whether a multi-dialect Vietnamese TTS model fine-tuned on the Fairytaler that Fakes Fluent and Faithful speech with Flow Matching (F5-TTS) architecture can accurately generate dialect-specific acoustic characteristics.

Literature Review & Methodology

Literature Review:

- **TTS Pipelines:** Traditional two-stage pipelines (acoustic model and vocoder) have been surpassed by modern flow-matching architectures. F5-TTS uses Optimal Transport Conditional Flow Matching (OT-CFM) with a Diffusion Transformer (DiT) backbone, improving training efficiency and inference speed.
- **F5-TTS**: Builds on E2 TTS by adding ConvNeXt text refinement and Sway Sampling at inference, reducing required steps while preserving audio quality. Its alignment-free design allows it to learn natural speech rhythms organically.
- **Vietnamese Dialects:** The Northern, Central, and Southern Vietnamese dialects differ in consonants, vowels, and tones. Tones are lexically contrastive, making accurate tonal synthesis critical.
- **Prior Work:** Pooling different dialects alone has not improved ASR, suggesting that dialect variation is a persistent challenge. It is unknown whether multi-dialect TTS training preserves regional characteristics or converges toward the dominant Northern dialect.

Methodology:

- **Baseline Model:** No task-specific baseline; F5-TTS was pre-trained on English and Mandarin.
- **Dataset:** Vietnamese Multi-Dialect (ViMD) dataset; approximately 6-7 hours from 3 dialects were extracted for ~20 total hours of speech data. All speech data is 24 kHz.
- **Speech-to-Text Evaluation:** The generated F5-TTS speech will be fed to a Vietnamese ASR model (Chunkformer-CTC-Large-Vie or PhoWhisper) and the ASR model's output text will be compared to the original text to determine accuracy.
- **Metrics**: Word Error Rate (WER), dialect classification accuracy between different models, dialect confusion matrix

Contribution:

- Vy Bui-Nguyen will work on parsing, cleaning, and extraction of data.
- Quenton Ni will work on fine-tuning the F5-TTS model for the task.
- Annalise Xiao will work on evaluating the model using predefined metrics.
- Cheston Opsasnick will work on documenting progress and generating the final report.

Experimental Results

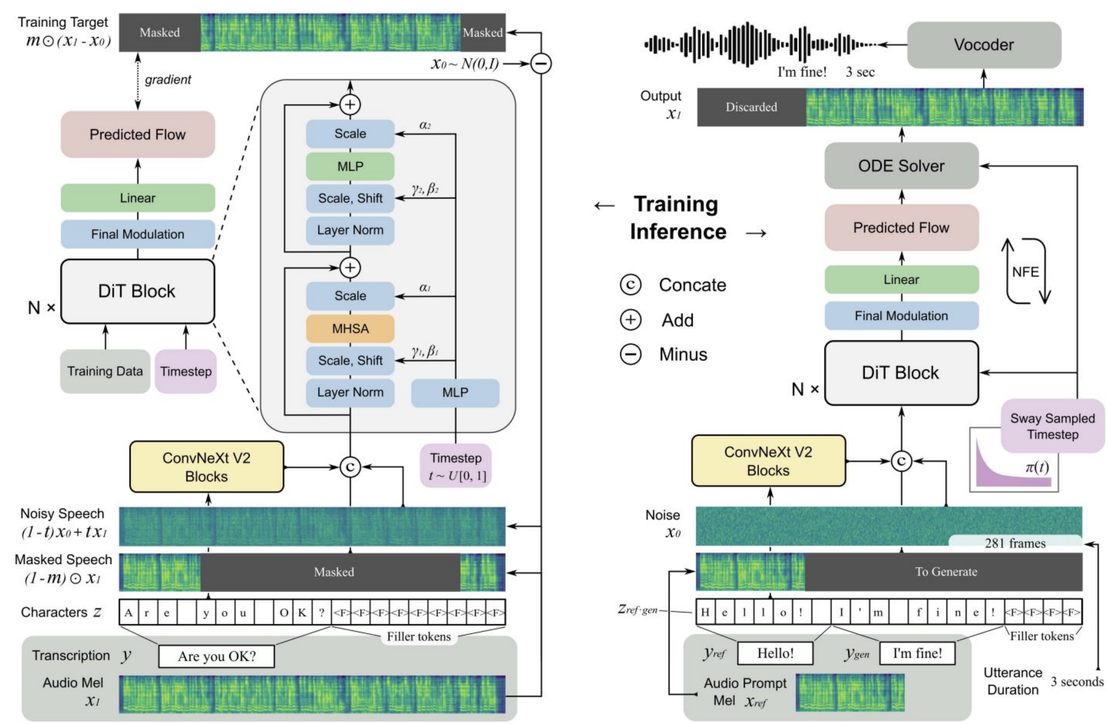

Figure 1: Overview of F5-TTS Training (left) and Inference (right)

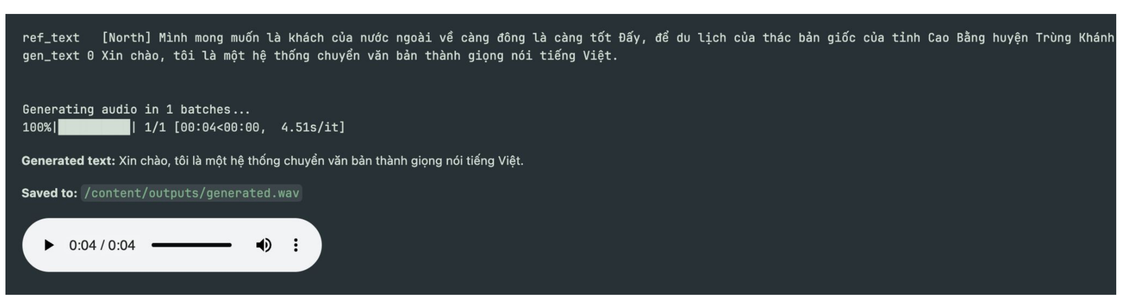

Figure 2: Northern Vietnamese Reference Text and Generated Output

Figure 3: Southern Vietnamese Reference Text and Generated Output

Discussion

Main Findings:

- Training on approximately 20 hours of speech data generated mostly unintelligible outputs characterized by short, incoherent phoneme sequences. While isolated Vietnamese words were occasionally produced, the majority of generated speech remains unrecognizable as fluent Vietnamese.
- Speaker tone was captured reasonably, suggesting that the model has begun learning high-level speech patterns.
- Dialect-specific embedding integration is ongoing; The project's current efforts are focused on achieving baseline Vietnamese intelligibility prior to starting evaluation for each dialect.

Plan for Final Report:

- Implement audio segmentation to identify optimal clip boundaries within the 15-second window, ensuring sufficient speech content is captured per sample.
- Develop transcript alignment logic to synchronize reference text with segmented audio clips, which will account for silence and inter-utterance gaps.
- Scale training data to approximately 50 hours of speech to generate intelligible Vietnamese speech, followed by full dialect-aware evaluation using a Vietnamese Automatic Speech Recognition (ASR) model.
- Analyze whether the F5-TTS model converges toward Northern Vietnamese dialect.
- Compile final per-dialect Word Error Rate (WER) scores into tables and incorporate cross-model comparison figures for the final report
- Update and complete project webpage with final results, figures, and generated audio samples.

Limitations

- **Constrained training data:** Only a subset of the ViMD dataset (total ~102 hours) was used, which may limit model generalization. Standard TTS models typically require 50 to 100+ hours minimum to achieve intelligible output, and multi-dialect models need more.
- **Bias towards Northern dialect**: The ViMD dataset is more skewed towards the Northern Vietnamese dialect.
- **Central dialect data scarcity:** Central Vietnamese is the least represented dialect in publicly available Vietnamese speech data.
- **No subjective evaluation:** No Mean Opinion Score (MOS) or human listening measures are included.

Key References

[1] Ben Pham and Sharynne McLeod. 2016. Consonants, vowels and tones across vietnamese dialects. International Journal of Speech-Language Pathology, 18(2):122–134.

[2] Nguyen Van Dinh, Thanh Chi Dang, Luan Thanh Nguyen, and Kiet Van Nguyen. 2024. Multi-dialect Vietnamese: Task, dataset, baseline models and challenges. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 7476–7498, Miami, Florida, USA. Association for Computational Linguistics.

[3] Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, Jian Zhao, Kai Yu, and Xie Chen. 2024. F5-tts: A fairytaler that fakes fluent and
faithful speech with flow matching. arXiv preprint arXiv:2410.06885.